

NIRMAL PATEL

Software Engineer at Intel

EXECUTIVE SUMMARY

Software engineer seeking new challenges and opportunities in the fields software development, embedded systems, and operating systems to utilize my hardware-software skills to contribute to the organization.

SKILLS

- Programming/ OS : C, C++, shell, python, Linux upstream kernel, OpenBMC
- Hardware : Intel VMD, Intel 3DXP Optane memory, Intel cable modem
- Tools/ Software : git, jira, IXIA traffic simulator, CD Router, Jsystem, CE Router, Klockwork, redfish, Bugzilla, Protex & Coverity for static code and binary analysis
- Protocols/ standards : PCI, TCP/IP, UDP, ICMP, IPv4/v6, DHCP, DNS, I2C

WORK EXPERIENCE

Intel Corporation

Chandler, AZ

Driver Development Engineer

(October 2020 – Present)

As a technical lead of Linux VMD driver team, my role includes feature development, requirements analysis and scoping, being the first point of contact for validation and customer issues related to Intel's VMD (volume management device) driver for Linux.

- Linux kernel maintainer for VMD PCI codebase; as a maintainer and a Linux community member, I am responsible for upstreaming new VMD changes and bugfixes to enable all Linux OS vendors to backport them.
- Owner of VMD replacement driver; replacement driver is a backported VMD kernel driver that includes performance features and bugfixes which are not part of upstream kernel but requested by customer.
- Create security threat models, collect compliance evidence, and present them to Intel's security forum to obtain SDL compliance for product release as an SDL security expert
- Participate in weekly debug meetings with validation and customer facing engineers to discuss new and present issues, prioritize them at program level, and provide suggestions to maintain a low number of outstanding issues.
- Lead weekly bug scrub meetings to assign tasks to team members based on priority and complexity.

Software Engineer

(January 2019 – October 2020)

Sharing ownership of the management software for Intel Optane memory with the team; and responsible for developing and supporting features for memory manageability and configurations. This open-source software stack enables OS and UEFI to talk to the module's firmware, generate logging information, update the firmware, and configure operating modes.

Feature development

- Leading and owning a proof of concept of managing memory modules out of band by Redfish in OpenBMC project for datacenter server management. Redfish uses Dbus to obtain user requested data from the registered service, and this service communicates to the memory controller using i2c bus.
- Design necessary test scenarios for system diagnostics and implement the tests. These tests involve obtaining information from firmware and platform configuration data (PCD) modules and verifying the results to generate required events with logs.

Software Engineer

(December 2015 - December 2018)

As a part of the gateway software engineering team, responsibilities included developing features related to network stack, providing customer support, and testing performance of cable modem.

Feature development

- Implement multi-switch solution for Yocto based RDKB-OS, enabling cable modem to support multiple Intel Ethernet switches. These code changes involved adding a hardware abstraction layer (HAL) between applications and driver. These changes gave the cable modem an ability to raise LAN aggregated throughput from 2.5 Gbps to 5 Gbps using LAG (Link Aggregation).

- Develop the scripts to allow Linux to identify and start two different types of network hardware on the same reference board. These scripts are triggered by systemd during boot up, and the script identifies whether PHY or Ethernet switch is connected.
- Found solution to locate and to fix multiple memory leaks in open-source components by integrating the run time memory checker tool into Intel's SDK, allowing the release of better quality software to the customer.

Customer support and security

- Root-cause the bottleneck in performance on customer's reference platform and implement the solution of load balancing between CPUs and firewall optimization to help the reference platform reach 930 Mbps downstream throughput. Firewall optimization is done by profiling which function was being processed the maximum number of times; thus, changing the algorithm helped achieve better bandwidth.
- Collaborate with DOS (Denial of Service) team in a very time sensitive activity of adding firewall protection and rate-limit rules for specific UDP and TCP ports to protect the interfaces against DOS attack.

System validation

- Lead the activity of transitioning automation in system validation. More than 350 test cases, involving regression testing, unit testing, and performance testing, are done by Jsystem automation tool.

SDL

- Achieve quarterly SDL compliance goals for Intel's software by cleaning up Intel owned patches on open-source components. Remove unused binaries and utilities; upstream patches to RDKB to reduce Intel's footprint in changing open-source code; close unused ports.

Volteo LLC

Intel Chandler, AZ

Intern IOT Engineer

(June 2015 - August 2015)

- Work with various sensors, Intel Edison embedded reference board, and gateways for ASU smart stadium initiative.
- Develop python script that automatically flashes image on reference platform using serial bus.

Tata Technologies Ltd

Pune, India

Graduate Engineer

(January 2013 - July 2013)

- Worked as client support engineer in manufacturing automation department.
- Technology used: i.e. MySQL, RSLogix 5000, RSLogix500, cimplicity, Beijer 2.0.

EDUCATION

- **Master of Science - Computer Engineering** Arizona State University, Tempe GPA: 3.5/ 4.0
- **Bachelor of Engineering - Electrical Engineering** Maharaja Sayajirao University, India GPA: 3.8/ 4.0

RELEVANT COURSEWORK

Computer Architecture	Hardware System Design	Embedded systems
Distributed & Multiprocessor OS	OS Internals	Software quality and assurance

ACADEMIC PROJECTS

User level thread library generation

(Spring 2015)

- Develop thread library that supports priority based preemptive scheduling of user level threads.
- User level threads implement functionality of priority inheritance and stack-based priority ceiling protocol.

Client-server application design and implementation in multiprocessors

(Fall 2014)

- Designed code for 3 client and one server. Data communication mechanism.
- Implement user level thread with TCB and Run Queue to get optimal performance with minimum waiting time. Perform validation using various locking, scheduling, and synchronization mechanisms like semaphores, mutex, and monitors on user threads.